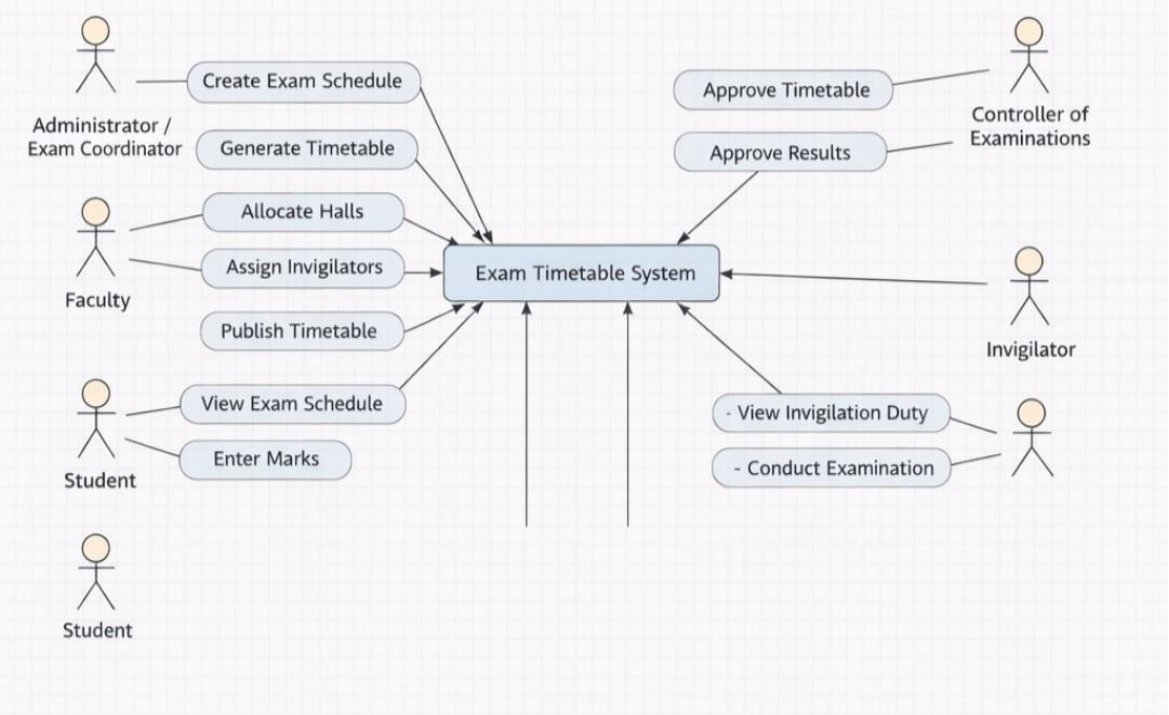
UML DIAGRAMS - UNIVERSITY EXAMINATION PROCESS

Use case



Explanation for Use Case Diagram

The **Use Case Diagram** shows **how different users (actors) interact with the exam timetable system** and what actions they can perform.

Actors in the System

1. Administrator / Exam Coordinator

The administrator is responsible for **managing the overall exam process**. Their main tasks include:

- Creating the exam timetable
- Generating the timetable
- Publishing the timetable
- Managing exam information

2. Faculty

Faculty members assist in organizing the exam by:

- Allocating exam halls
- Assigning invigilators
- Evaluating answer scripts
- Entering marks

3. Controller of Examinations
The controller ensures that the exam process is **properly monitored and authorized**. Their responsibilities include:

- Approving the exam timetable
- Approving final results before publication

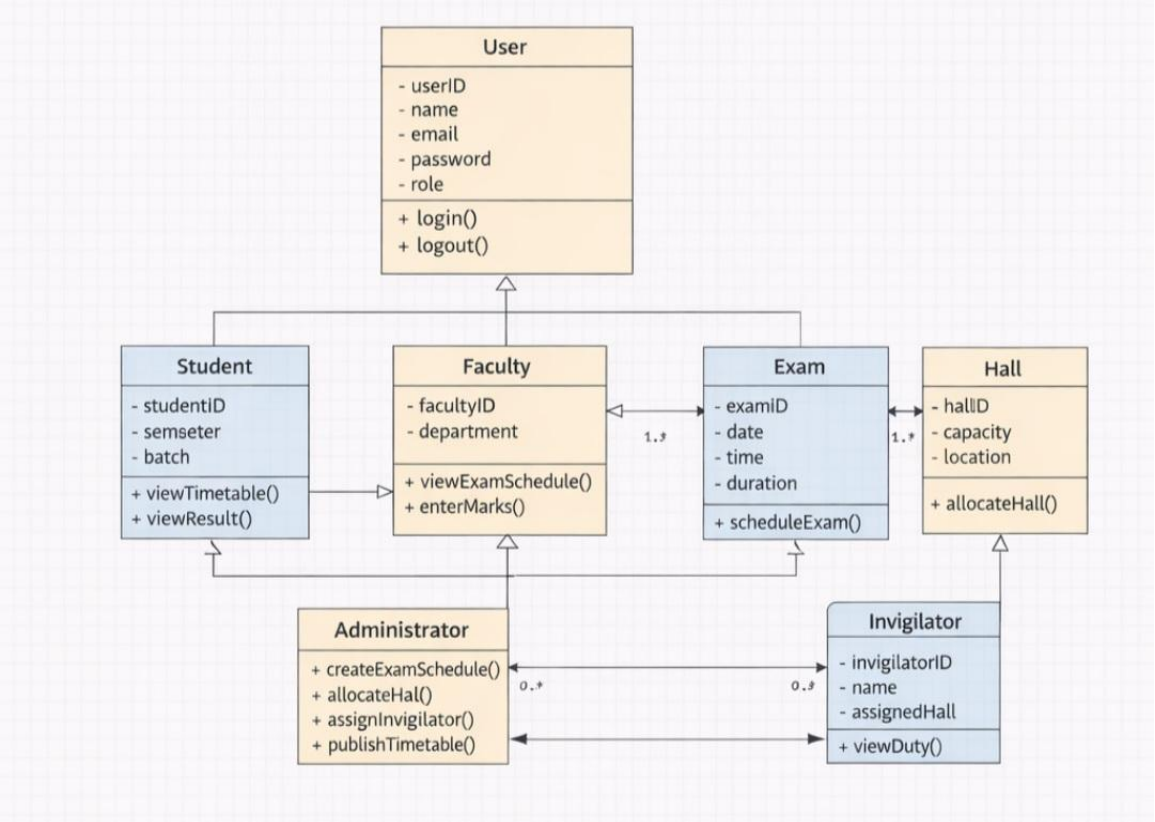
4. Invigilator
Invigilators supervise the exam during the exam session. Their tasks include:

- Conducting the examination
- Monitoring students
- Ensuring exam rules are followed

5. Student
Students interact with the system mainly to:

- View the exam timetable
- View exam results
- Apply for revaluation if needed

Class Diagram



Explanation of the Class Diagram

The **Class Diagram** shows the **structure of the examination system**. It explains the different **classes in the system, their attributes (data), and the relationships between them**.
It helps us understand **how the system is designed internally and how different objects interact with each other**.

Main Classes in the System

1. User Class (Base Class)

The **User class** is the main parent class in the system.
It contains common information that is shared by all types of users.

Some common attributes include:

- **User ID** – Unique identification for each user
- **Name** – Name of the user
- **Email** – Contact email
- **Login credentials** – Username and password for system access

Other classes such as **Student, Faculty, Administrator, and Invigilator** inherit these common properties from the **User class**.

2. Student Class

The **Student class** represents the students who appear for examinations.

Important attributes include:

- **Student ID**
- **Course or program details**
- **Exam results**

Students mainly interact with the system to:

- View the **exam timetable**
- View their **exam results**
- Request **revaluation if needed**

3. Faculty Class

The **Faculty class** represents teachers who are responsible for academic activities in the examination system.

Their responsibilities include:

- Preparing **question papers**
- **Evaluating answer scripts**
- **Entering marks** into the system

- Sometimes supervising exams as invigilators

4. Administrator Class

The **Administrator class** represents the exam authority who manages the entire examination process.

The administrator performs tasks such as:

- Creating the **exam timetable**
- Managing **exam schedules**
- Coordinating **exam halls and invigilators**
- Managing overall exam activities in the system

5. Invigilator Class

The **Invigilator class** represents staff members who supervise students during the exam.

Their responsibilities include:

- Monitoring students during the exam
- Distributing question papers
- Collecting answer scripts after the exam

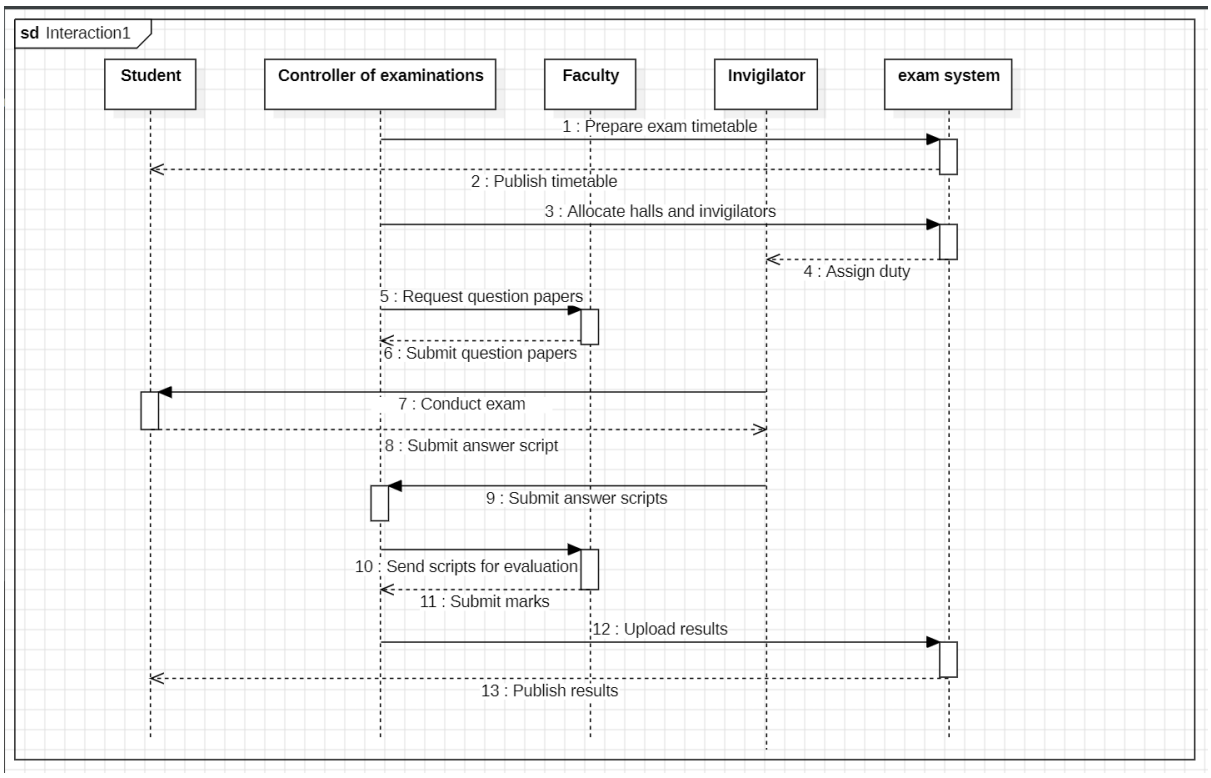
Relationships Between Classes

The class diagram also shows how different classes are connected.

- An **Exam** can be conducted in **one or more halls**.
- A **Hall** can host **different exams at different times**.
- The **Administrator** manages exams and assigns invigilators.
- **Faculty members** evaluate answer scripts and enter marks into the system.
- **Students** are associated with exams because they appear for them and receive results.

These relationships help the system manage the entire examination process efficiently.

SEQUENCE DIAGRAM



Explanation of the Sequence Diagram – University Examination System

The sequence diagram illustrates the **interaction between different participants involved in the university examination process**. It shows the **order in which activities occur from exam preparation to result announcement**.

Participants

The main participants involved in the system are:

- **Student** – Appears for the examination and receives results.
- **Controller of Examinations (COE)** – Manages and oversees the entire examination process.
- **Faculty** – Prepares question papers and evaluates answer scripts.
- **Invigilator** – Supervises the examination and collects answer scripts.
- **Exam System** – The software system that manages scheduling, assignments, and result publication.

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1. Exam Timetable Preparation

The process begins when the **Controller of Examinations prepares the examination timetable** using the **Exam System**.
The system processes this request and **publishes the timetable**, making it available to students.

2. Hall Allocation and Invigilator Assignment

Next, the **Controller of Examinations allocates examination halls and assigns invigilators** through the system.
The **Exam System sends duty assignments to invigilators**, informing them about their responsibilities.

3. Question Paper Preparation

The **Controller of Examinations requests question papers from the faculty** members responsible for each subject.
Faculty members prepare the question papers and **securely submit them to the Controller of Examinations**.

4. Conducting the Examination

On the examination day, the **invigilator conducts the exam in the assigned hall**.
The invigilator distributes the question papers to students and supervises the examination process.
After completing the exam, **students submit their answer scripts to the invigilator**.

5. Collection of Answer Scripts

Once the examination ends, the **invigilator collects all answer scripts** and submits them to the **Controller of Examinations** for further processing.

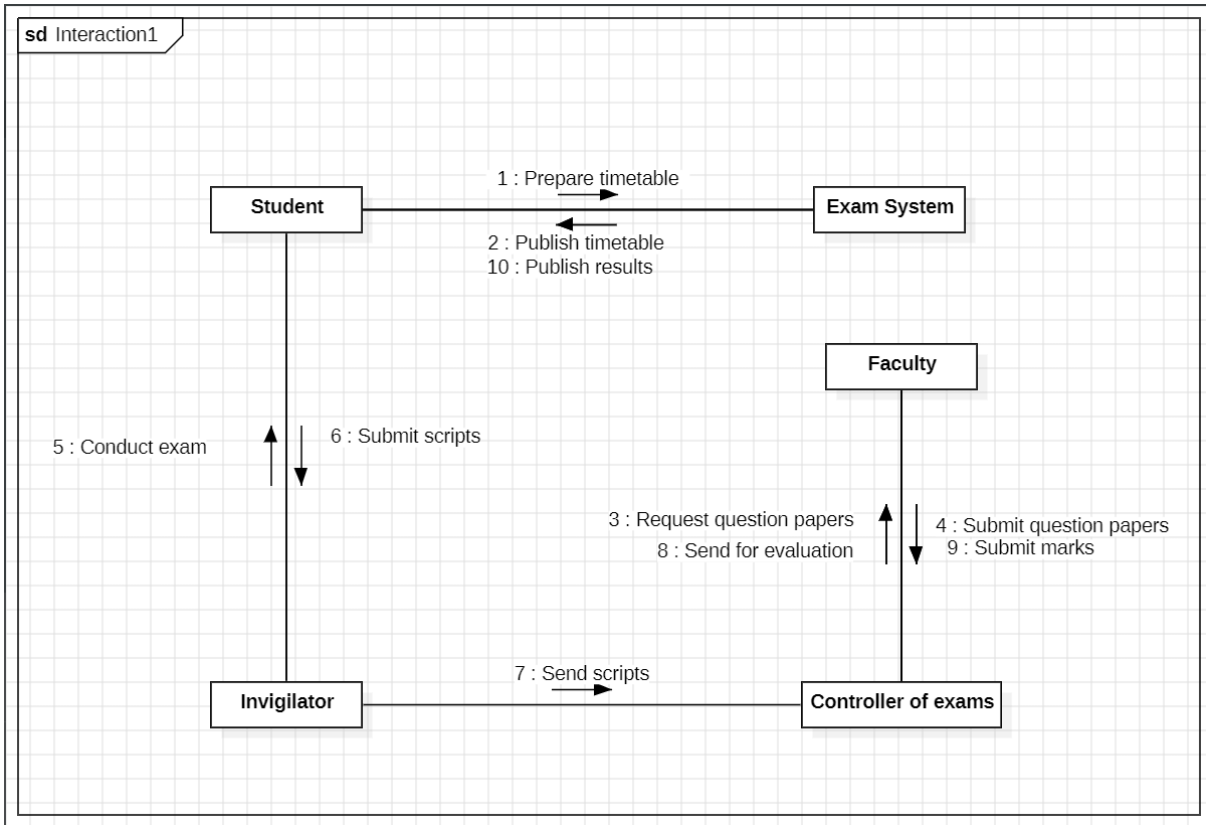
6. Evaluation of Answer Scripts

The **Controller of Examinations sends the answer scripts to the respective faculty members** for evaluation.
Faculty members **evaluate the scripts and submit the marks back to the Controller of Examinations**.

7. Result Processing and Announcement

Finally, the **Controller of Examinations uploads the marks into the Exam System**.
The system processes the results and **publishes them for students to view**.

COLLABORATION DIAGRAM



Explanation of the Communication (Collaboration) Diagram

The communication diagram illustrates the **interaction and communication between different participants involved in the university examination process**. It focuses on **how objects are connected and how messages are exchanged between them** to complete the examination workflow. The **numbers on the messages indicate the sequence in which the interactions occur**.

Participants

The main participants involved in the system are:

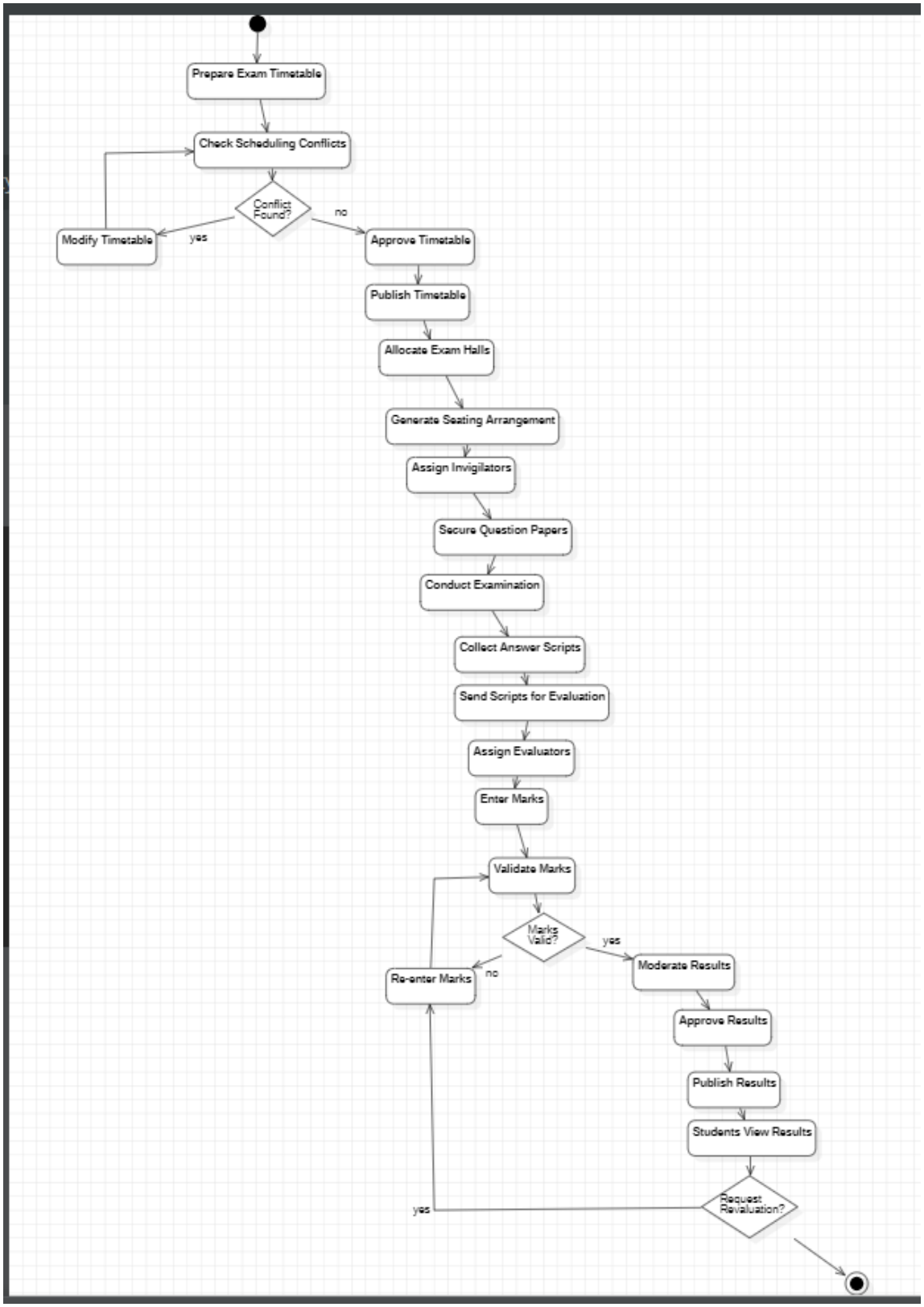
- **Controller of Examinations (COE)** – Responsible for managing and supervising the entire examination process.
- **Exam System** – The software system that supports scheduling, management, and result publication.
- **Faculty** – Responsible for preparing question papers and evaluating answer scripts.
- **Invigilator** – Conducts the exam in the examination hall and collects answer scripts.
- **Student** – Appears for the examination and receives the results.

Workflow Description

- The process begins when the Controller of Examinations prepares the exam timetable using the Exam System.

- The Exam System publishes the timetable, allowing students to view their exam schedule.
- The Controller of Examinations requests question papers from the faculty members responsible for different subjects.
- Faculty members prepare and submit the question papers to the Controller of Examinations.
- During the examination, the invigilator conducts the exam and supervises the students in the examination hall.
- After completing the exam, students submit their answer scripts to the invigilator.
- The invigilator collects the answer scripts and sends them to the Controller of Examinations.
- The Controller of Examinations forwards the answer scripts to faculty members for evaluation.
- Faculty evaluate the answer scripts and submit the marks.
- Finally, the Exam System publishes the results, allowing students to view their exam performance.

Activity diagram



Explanation for Activity Diagram

The **Activity Diagram** represents the **workflow of the entire examination management process** from preparing the exam timetable to publishing the final results. It shows the **sequence of activities and decision points** in the system.

1. Prepare Exam Timetable
The process begins when the **exam authority or administrator prepares the exam timetable**. This includes deciding the exam dates, times, and subjects.

2. Check Scheduling Conflicts
After the timetable is prepared, the system checks for **scheduling conflicts**, such as:

Two exams scheduled at the same time

Students having overlapping exams

Limited hall availability

If a conflict is detected, the timetable must be **modified**.
If no conflict exists, the timetable moves to the **approval stage**.

3. Publish Timetable
Once the timetable is finalized, it is **published in the system** so that students and faculty members can access it.

4. Allocate Exam Halls
The system assigns **specific halls or classrooms** for each exam based on capacity and availability.

5. Generate Seating Arrangement
After halls are assigned, the system generates **seating arrangements for students**, ensuring that students are distributed properly across halls.

6. Assign Invigilators
Faculty members are assigned as **invigilators** to supervise the exam in different halls.

7. Secure Question Papers
Question papers are prepared and **securely distributed to the exam halls** to ensure confidentiality.

8. Conduct Examination
The examination is conducted where **students write the exam under the supervision of invigilators**.

9. Collect Answer Scripts
After the exam ends, all **answer sheets are collected** from the students.

10. Send Scripts for Evaluation
The collected scripts are **sent to teachers or evaluators** for checking.

11. Assign Evaluators and Enter Marks
Evaluators check the answer scripts and **enter the obtained marks into the system**.

12. Validate Marks
The system verifies whether the entered marks are correct and complete.
If an error occurs, the evaluator must re-enter the marks.

13. Moderate and Approve Results
The controller of examinations may **review or moderate the marks** before approving the final results.

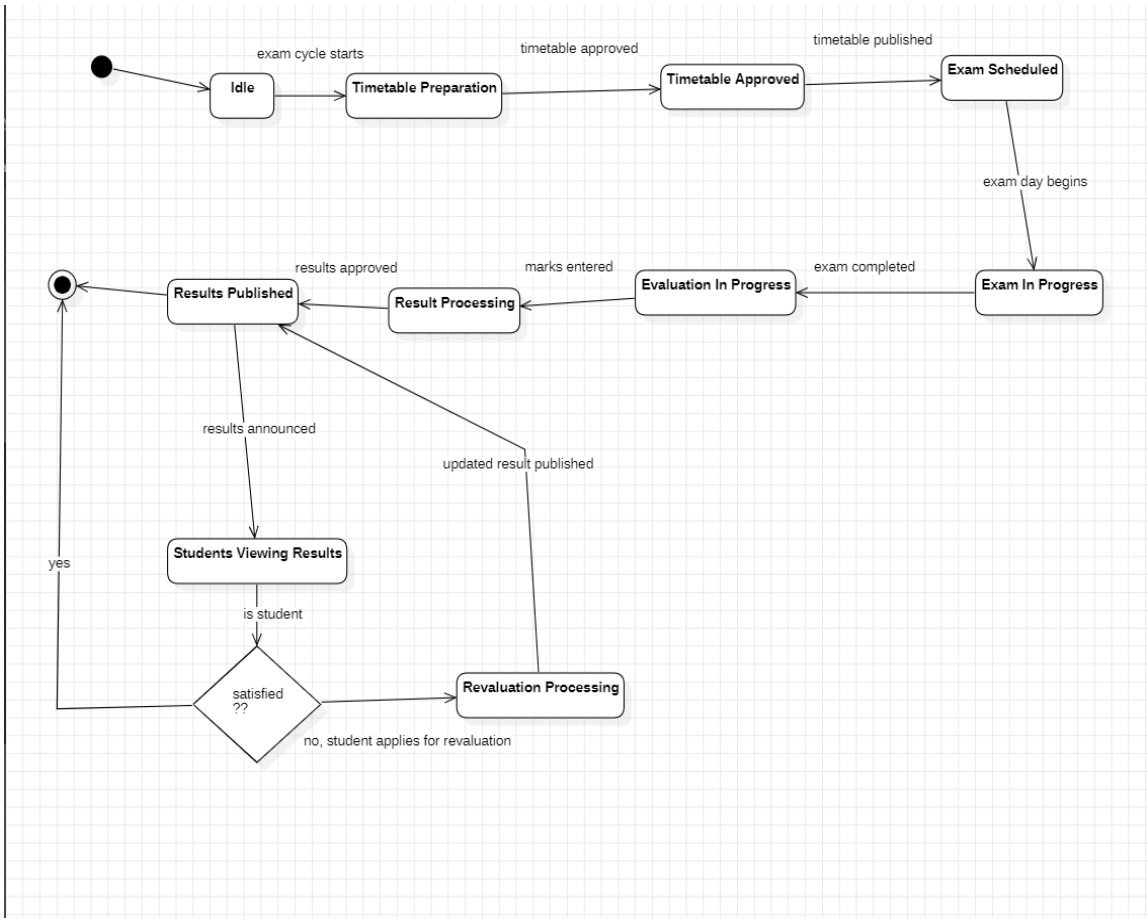
14. Publish Results
Once approved, the results are **published for students**.

15. Students View Results
Students log into the system to **view their exam results**.

If a student is not satisfied with the marks, they can **request revaluation**, where the answer script is checked again.

Finally, once the revaluation process is completed and results are finalized, the **exam process ends**.

Statechart



Explanation for Statechart Diagram

The **Statechart Diagram (State Machine Diagram)** represents the **different states of the exam system and how it transitions from one state to another** during the examination process.

States in the System

1. Idle State

Initially, the system is in the **Idle state**, meaning no exam process is currently active.

2. Timetable Preparation

The exam process begins when the system moves to the **Timetable Preparation state**, where the exam schedule is created.

3. Timetable Approved

After the timetable is prepared, it is **reviewed and approved by authorities**, such as the Controller of Examinations.

4. Exam Scheduled

Once approved, the timetable is **published**, and the exams become officially scheduled for students.

5. Exam in Progress
When the exam date arrives, the system transitions to the **Exam in Progress state**, where students write their examinations.

6. Evaluation in Progress
After the exam is completed, answer scripts are sent to teachers, and the system enters the **Evaluation in Progress state**, where papers are checked and marks are entered.

7. Result Processing
Next, the system processes the marks in the **Result Processing state**, which includes verifying marks and approving the results.

8. Results Published
After approval, the system moves to the **Results Published state**, allowing students to view their results.

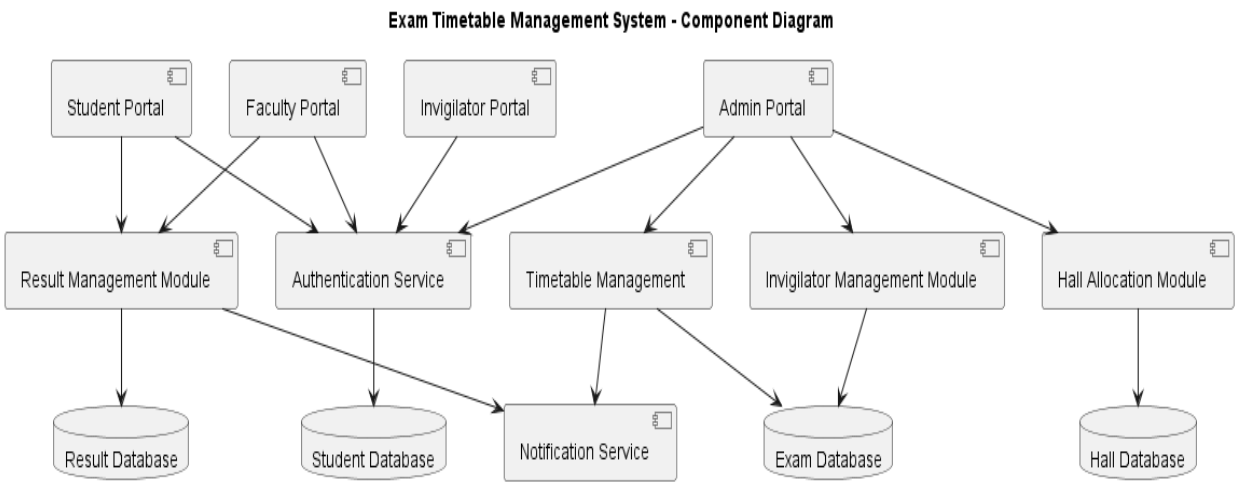
9. Student Decision State
Students review their marks and decide whether they are satisfied.

- If **satisfied**, the process ends.
- If **not satisfied**, the system moves to **Revaluation Processing**.

10. Revaluation Processing
In this state, the student's answer script is **rechecked**, and marks may be updated.

11. Final Results Published
The updated results are published, and the **exam lifecycle is completed**.

Component Diagram



The main components involved in the system are:

User Interface Components – These components allow different users to interact with the system.

- **Student Portal** – Enables students to view exam timetables and results.
- **Faculty Portal** – Allows faculty members to view exam schedules and submit marks.
- **Admin Portal** – Used by the Controller of Examinations to manage exams and scheduling.
- **Invigilator Portal** – Allows invigilators to view their assigned exam duties.

Authentication Service – Responsible for validating user credentials and controlling secure access to the system.

Timetable Management Component – Handles the creation, modification, and publication of examination schedules.

Hall Allocation Component – Manages the assignment of examination halls based on capacity and availability.

Invigilator Management Component – Assigns invigilators to examination halls and manages invigilation duties.

Result Management Component – Handles the entry of marks, calculation of grades, and publication of final results.

Notification Service – Sends notifications such as exam schedule announcements and result updates to users.

Database Components – Store system data including student records, exam schedules, hall information, and results.

Workflow Description

The process begins when users access the system through their respective portals such as the Student Portal, Faculty Portal, Admin Portal, or Invigilator Portal.

All users must first authenticate themselves through the **Authentication Service** before accessing system functionalities.

The **Admin Portal**, managed by the Controller of Examinations, interacts with the **Timetable Management Component** to create and publish examination schedules.

The **Hall Allocation Component** assigns examination halls based on the number of students and available space.

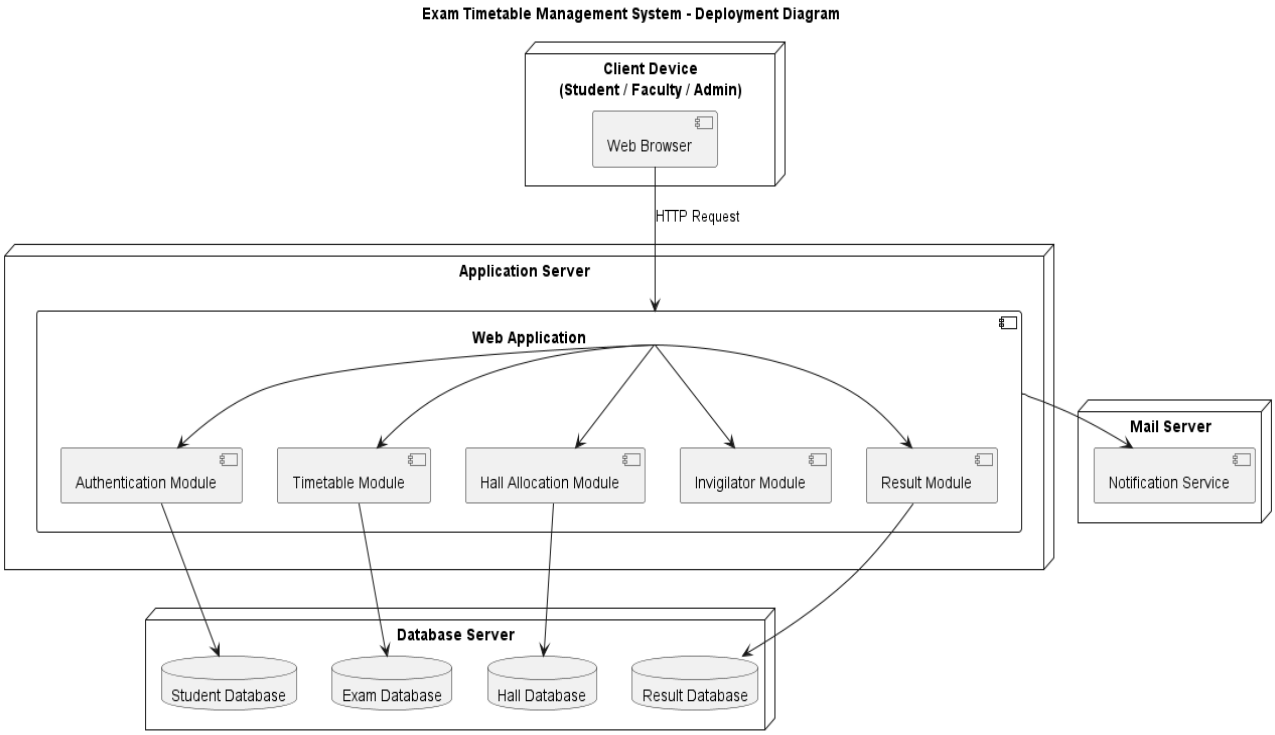
The **Invigilator Management Component** assigns invigilators to examination halls to supervise examinations.

Faculty members interact with the **Result Management Component** to enter and submit marks after evaluating answer scripts.

The **Result Management Component** calculates grades and publishes the results through the system.

Finally, the **Notification Service** informs students and faculty members about exam schedules, updates, and result announcements.

Deployment Diagram



The **Deployment Diagram** represents the physical architecture of the University Examination System. It shows how software components are deployed on hardware nodes and how these nodes communicate with each other during system operation. This diagram helps visualize the distribution of the system across different servers and devices.

Nodes

The main nodes involved in the system are:

Client Device – Represents the devices used by students, faculty members, invigilators, and administrators to access the system. These devices typically use web browsers or mobile applications to interact with the examination system.

Web Server / Application Server – Hosts the University Examination System application. It processes user requests, executes business logic, and communicates with the database server.

Database Server – Stores all the data related to the examination system including student details, exam schedules, hall allocations, invigilator assignments, and examination results.

Mail or Notification Server – Responsible for sending notifications such as exam timetable announcements and result updates to users.

Workflow Description

The process begins when users such as students, faculty members, invigilators, or administrators access the system using their client devices.

These client devices send requests to the **Web Server** through the internet using web browsers or applications.

The **Web Server** processes the requests and executes the required operations such as exam scheduling, hall allocation, invigilator assignment, or result processing.

To retrieve or store information, the web server communicates with the **Database Server**, where all examination-related data is maintained.

If notifications are required, the system interacts with the **Mail or Notification Server** to send updates regarding exam schedules, duties, or results.

Finally, the processed information is sent back to the client devices, allowing users to view schedules, duties, or results through the system interface.